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Marks	0/11
Grade	0 out of 100

Question 1

Not answered

Marked out of 1

In a study examining the relationship between domestic abuse and depression, women who do and do not suffer abuse are followed up over a five-year period. It is found that 70 of the 500 individuals who undergo domestic abuse develop depression; 150 of 3000 individuals who don't get abused develop depression. This study is best described as

Select one:

- ☐ Ecological study
- ☐ Cross sectional survey
- ☐ Case control study
- ☐ Cohort study
- ☐ Controlled trial

Your answer is incorrect.

This is a follow-up in a prospective manner where exposure is determined at the outset. This is a cohort study.

The correct answer is: Cohort study

Question 2

Not answered

Marked out of 1

A cross-sectional comparison of the incidence of schizophrenia at three periods 1881-1902, 1978-1980 and 1992-1994 was performed using hospital admission records at Mapperley Hospital, Nottingham. Research diagnostic criteria were applied retrospectively to determine the diagnosis. It was concluded that no significant changes in the incidence of schizophrenia occurred across the 114 years considered in the study. Which of the following, if present, can invalidate such a conclusion?

Select one:

- ☐ Lack of treatment data
- ☐ Lack of appropriate denominators
- ☐ Lack of randomisation
- ☐ Lack of adequate follow-up
- ☐ Lack of control group

Your answer is incorrect.

Any study comparing incidence rates must have a specified denominator; in this case it will be mid-period population during the observed years. This study is a retrospective survey - we do not need randomisation or interventions to answer the research question.

The correct answer is: Lack of appropriate denominators

Question 3

Not answered

Marked out of 1

National Comorbidity Survey examined a possible association between traits of neuroticism and lifetime prevalence of various psychiatric disorders. Trained non-clinical staff screened randomly chosen individuals using well-validated questionnaires to establish lifetime diagnoses in addition to collecting data on drug use, alcohol intake and various personality traits. What type of study is this?

Select one:

- ☐ Retrospective cohort study
- ☐ Cross-sectional
- ☐ Case-control
- ☐ Prospective cohort study
- ☐ Clinical trial

Your answer is incorrect.

Most epidemiological studies are cross-sectional surveys where point prevalence and/or lifetime prevalence are estimated.

The correct answer is: Cross-sectional

Question 4

Not answered

Marked out of 1

A doctor notices three cases of new onset myoclonic epilepsy in his inpatient unit. Accidentally, he discovers that all three patients had been weaned off the same depot medication two weeks ago. To alert the wider medical community about a possible discontinuation effect of this drug, the most preferred method of publishing this observation is

Select one:

- ☐ Case series
- ☐ Meta analysis
- ☐ RCT
- ☐ N of 1 trial
- ☐ Systematic review

Your answer is incorrect.

Case reports are based on the experience of only one person; that is, the information is anecdotal. Case series lack an appropriate comparison group. But despite these limitations, they need much less time to write up and publish compared to other research designs.

The correct answer is: Case series

Question 5

Not answered

Marked out of 1

In which of the following studies exposure data are available for groups but not for their individual members?

Select one:

- ☐ Cross sectional surveys
- ☐ Cross over trials
- ☐ Cluster trials
- ☐ Multicentre surveys
- ☐ Ecological study

Your answer is incorrect.

Ecological studies use data from whole populations rather than individuals, they cannot link an exposure to a disease occurrence in the same person. Another limitation is the inability to control for the effects of other factors that may be associated with the outcome.

The correct answer is: Ecological study

Question 6

Not answered

Marked out of 1

Which of the following correctly describes ecological fallacy

Select one:

- ☐ Erroneous result due to selecting smaller groups instead of larger groups
- ☐ Erroneous result due to selecting larger groups instead of smaller groups
- ☐ Ascribing the characteristics of a group to all individuals in that group
- ☐ Failure to examine temporal relationships between exposures and outcomes
- ☐ Failure to examine dose response relationship while claiming causal association

Your answer is incorrect.

An ecological fallacy is "an error in the interpretation of statistical data in an ecological study, whereby inferences about the nature of specific individuals are based solely upon aggregate statistics collected for the group to which those individuals belong" (http://wiki.gis.com/wiki/index.php/Ecological_fallacy).

The correct answer is: Ascribing the characteristics of a group to all individuals in that group

Question 7

Not answered

Marked out of 1

Retrospective cohort studies are characterized by all of the following except

Select one:

- ☐ The required sample size is smaller than that needed for a prospective cohort study
- ☐ Incidence rates may be computed
- ☐ They are useful for rare exposures
- ☐ The study groups are exposed and nonexposed
- ☐ The required sample size is similar to that needed for a prospective cohort study

Your answer is incorrect.

The only difference from prospective study is in calendar timings of data collection and, as a result, the requirement for case registers/records to tease out data when doing a retrospective cohort study

The correct answer is: The required sample size is smaller than that needed for a prospective cohort study

Question 8

Not answered

Marked out of 1

Which of the following is NOT an advantage of a prospective cohort study?

Select one:

- ☐ Incidence rates can be calculated
- ☐ Recall bias is minimized compared with a case-control study
- ☐ Precise measurement of exposure is possible
- ☐ It usually costs less than a case-control study
- ☐ Many disease outcomes can be studied simultaneously

Your answer is incorrect.

Time resources, manpower and monetary costs are higher though cohort studies may offer more value for research investment when compared to multiple case-control studies

The correct answer is: It usually costs less than a case-control study

Question 9

Not answered

Marked out of 1

Which one of the following is a disadvantage of carrying out a cohort study?

Select one:

- ☐ Prospective design cannot be achieved
- ☐ Temporal association cannot be deduced
- ☐ For a disease with high incidence, large amount of subjects will be needed to complete the study
- ☐ Dose response effect cannot be identified
- ☐ It is time consuming

Your answer is incorrect.

Cohort studies are usually prospective; they are costly and time consuming, and some can span for many decades. The ability to ascertain the temporal correlation between exposure and an outcome is a major strength of cohort design. For diseases that occur at a high frequency on exposure, a relatively smaller cohort should be sufficient when compared to rare diseases.

The correct answer is: It is time consuming

Question 10

Not answered

Marked out of 1

Which of the following study design is best suited to evaluate properties of a screening tool?

Select one:

- ☐ Cohort
- ☐ Randomised trial
- ☐ Meta-analysis
- ☐ Cross sectional design
- ☐ Case control

Your answer is incorrect.

It is important to administer both gold standard and the evaluated tool to all study subjects. A cross sectional design is often used for this purpose.

The correct answer is: Cross sectional design

Question 11

Not answered

Marked out of 1

What type of study design can be used to find out the number of individuals with mental illness among those who are dependent on alcohol?

Select one:

- ☐ Case control study
- ☐ Cohort study
- ☐ Cross sectional study
- ☐ Open trial
- ☐ Nested study

Your answer is incorrect.

Cross-sectional studies are well suited for prevalence estimates.

The correct answer is: Cross sectional study



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